

Carlos Fernando Revilla

Software Engineer | M.S. Data Science

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TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C# (.NET), Java, C++, SQL, SAS, R

Software Tools: AWS, Terraform, Docker, CI/CD, Apache Kafka & Spark, PostgreSQL, MongoDB, Solr, ETL/ELT Pipelines, TensorFlow, LLM Integration, Git, Jest

WORK EXPERIENCE

Philips | Full-Stack Software Engineer

Cambridge, MA | January 2026 - Present

- Architected a multi-service cloud platform on AWS ECS Fargate, using Terraform to provision API Gateway, ALB routing, and per-tenant state isolation, creating a scalable backbone capable of serving AI/ML model endpoints.
- Designed a fully automated CI/CD pipeline with GitHub Actions and OIDC-based AWS authentication, orchestrating Docker builds and zero-downtime deployments for multiple services.
- Supported in maintenance, and deployments of Quartz.NET background service to preserve data quality for a client-facing dashboard synced with a Citus/RDS that aggregates patient device data.
- Maintain and enhance an internal React/Vite dashboard used by clients to monitor real-time product data streams, leveraging modern AI-assisted development tools to accelerate feature delivery.

Meta | Data Labeling Analyst (Contract)

Remote | November 2025 – January 2026

- Annotated & Evaluated high-volume text datasets to train and optimize Meta generative machine learning models providing critical ground-truth data to improve algorithmic performance.
- Maintained a 98%+ quality assurance score by strictly adhering to complex, rapidly evolving content moderation and data labeling guidelines.
- Identified data trends and edge cases, collaborating with cross-functional quality control teams to refine labeling schemas and reduce annotation ambiguity.

Silverchair | Full-Stack Software Engineer

Charlottesville, VA | May 2024 - February 2025

- Diagnosed and resolved critical data integrity issues in a Solr pipeline by re-ingesting missing columns, restoring content accuracy and eliminating 28% of user-reported errors.
- Identified and resolved N+1 query bottlenecks against large SQL datasets through strategic caching and batching, improving search load times by ~300ms on web and mobile apps.
- Championed code quality by building a comprehensive Jest test suite, reducing manual QA efforts by 2 hours per week for data-critical features.

EDUCATION

M.S. - Data Science | University of Virginia

Expected May 2027

- Focus: Machine learning, Data engineering, Big Data Systems, and Natural Language Processing.

B.A. - Computer Science | University of Virginia

August 2020 - May 2024

- *Minor: Data Science - Relevant Courses:* Artificial Intelligence, Data Science Systems, Database Systems

Software Engineering Bootcamp | Codesmith, Full-time Immersive

May 2023 - September 2023

- *Focus: Advanced JavaScript, TypeScript, Database Design, State Management, React + Redux Toolkit*

PROJECTS

OS-Labs: KafkaNightOwl | Full-Stack Software Engineer

- Built a Message Data & Metrics Service using Node.js/Express, streamlining Kafka message transactions
- Designed a scalable React/TypeScript dashboard with Flux architecture, applying container patterns for maintainable UI state management.

Acne-AI | Full-Stack Software Engineer

- Fine-tuned Meta's DinoV2 foundation model for skin condition classification, achieving strong benchmark performance on a held-out test set using Google Colab's A100 GPUs.
- Integrated a React frontend with the Python model backend using Axios, optimizing image processing and reducing diagnosis latency.

TALKS & PUBLICATIONS

GraphQL & the N+1 Problem | Speaker

- Presented solutions to GraphQL's N+1 query issue. Talk sponsored by Codesmith, Jeeny, and Bractlet.